MPMC COURSE PROJECT

PostureSense Chair— Real-time Sensor Feedback & Personalised Correction

PosturePilot

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Introduction

- Poor sitting posture has become a major health concern in today's digital age, especially among students and professionals who spend long hours at desks.

- Prolonged slouching can lead to back pain, spinal deformities, and reduced productivity.

- To address this, our system aims to monitor sitting posture in real time using low-cost sensors and provide instant feedback and alerts to help users maintain healthy posture habits.

ROADMAP - ENGINEERING ASPECT

Phases

- **Phase-1:** Ideation and Research
- **Phase-2:** Prototype Making
- **Phase-3:** Testing and Iterations

Components Used

- **ESP32:** Processor/Brain
- **IR Sensor -** Measures distance to user's back HC-SR04

Working Logic

- **Calibration:** Record baseline distances.
- **Monitoring:** If seat shifts forward distance increases, assume bad posture
- After **delay,** triggers alert on app interface

Evolution of the Product

- **Past →** Passive braces & posture advice
- **Present →** Sensorized wearables with haptic feedback
- **Future →** ML-driven personalised coaching + clinical validation
- **Ou r Prototype →** A smart chair add-on that senses your sitting posture and gently reminds you to sit upright.

IEEE RESEARCH PAPER-1

WORKING LOGIC:

- Ultrasonic sensor detects posture by comparing measured distance with a preset threshold (6 cm).
- IMU sensor identifies posture deviations via angle changes in the thoracic spine region.
- ESP32 processes sensor data and sends it via Bluetooth to the mobile app.
- The Android app displays real-time posture data and issues alerts when improper posture is detected.

ELECTRONICS INVOLVED:

Core Controller:

- ESP32 microcontroller with built-in Wi-Fi/Bluetooth for wireless data transmission.

Sensors Used:

- HC-SR04 Ultrasonic Sensor mounted on the chair backrest to measure distance between user's back and chair.
- MPU-6050 IMU Sensor embedded in a T-shirt to monitor angular variations (pitch, roll, yaw) of the upper back.

Power Source:

- 3.7 V Li-ion rechargeable battery supplies the ESP32 and sensors.

Software Tools:

- Arduino IDE for programming ESP32; MIT App Inventor for developing the Android app.

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ELECTRONICS INVOLVED:

- ESP32S Wi-Fi module which sends processed posture data to the Arduino IoT Cloud.
- Accelerometer (ADXL335, 3-axis): Detects inclination of the user's spine.
- ADC0804 (8-bit): Converts analog accelerometer signals into digital data .
- 555 Timer (Astable mode): Generates clock frequency required by ADC for conversion.
- AT89S52 Microcontroller: Processes digital data , Compares readings and displays the result.

WORKING LOGIC:

- The AT89S52 microcontroller compares the reading with a preset threshold value (≈180).
- If value < threshold → normal posture (no alert). If value ≥ threshold → bad posture.
- For bad posture , Vibration motor is activated to alert the user.
- LCD (16×2) displays the digital value.
- Simultaneously, data is sent via ESP32 Wi-Fi module to the Arduino IoT Cloud.
- Mobile app/IoT dashboard shows real-time readings and graphs posture history.

Current Scenario

Most available posture correction devices are either:

- Too expensive and targeted toward medical use or
- Too basic, offering only vibration/buzzer alerts without data tracking.
- There's a lack of affordable, real-time, and user-friendly posture monitoring systems for everyday users.
- Hence, a simple, low-cost system integrating hardware feedback with a mobile app interface is highly relevant today.

Methodology & Working Principle

- The proposed system continuously monitors the user's sitting distance using an ultrasonic sensor fixed to a chair and controlled by an ESP32 microcontroller.

Working Steps:

1) **Distance Measurement**: Ultrasonic sensor measures the user's back distance in real time.

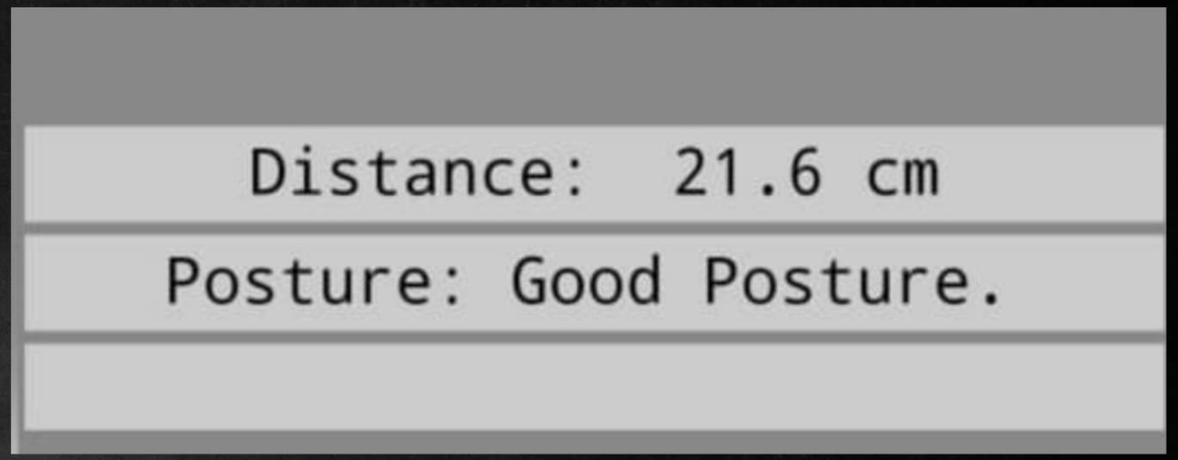
2) **Feedback Display:**

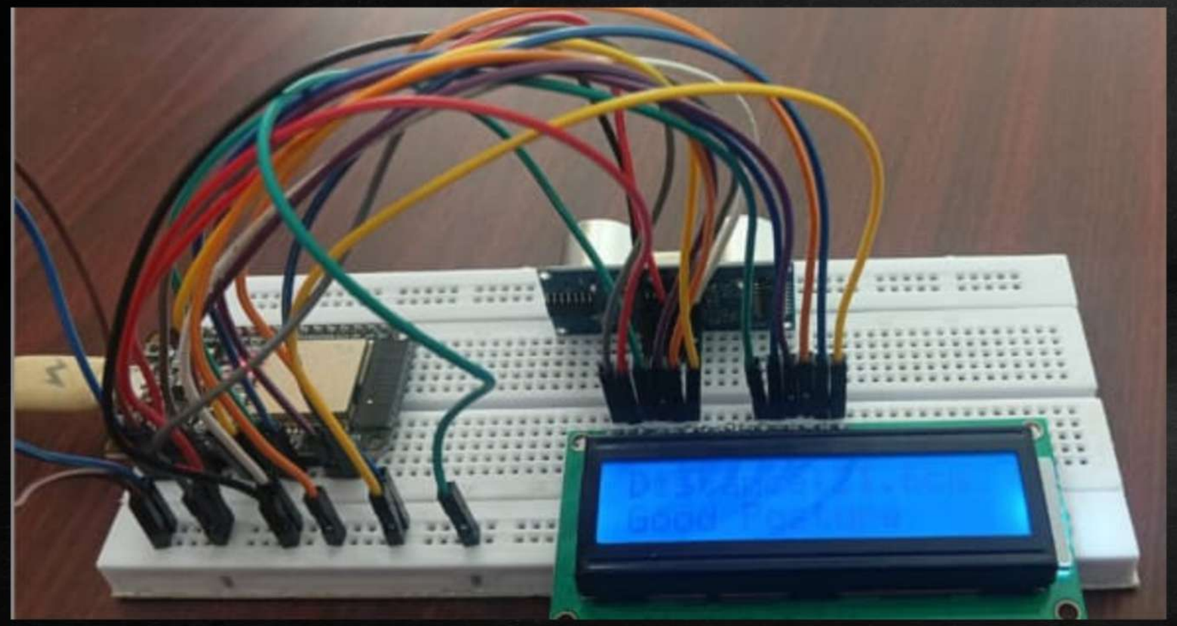
- LCD shows posture status as "Good" or "Bad."
- Bluetooth sends posture data to the MIT App Inventor mobile app.

3) **Timed Alerts:**

- If slouching continues for >10 seconds, a reminder is triggered.
- If sitting continuously for >45 minutes, the system alerts the user to take a break.

RESULTS OBTAINED





https://screenapp.io/app/v/xKcn4iP6iV

Future Enhancements

- **Cloud Integration:** Store posture data for long-term tracking and analytics.
- **Machine Learning:** Personalize feedback based on user behavior and seating patterns.
- **Multi-Sensor Setup:** Use IMU/gyroscope sensors for full-body posture analysis.
- **Vibration Feedback:** Add gentle haptic alerts instead of only visual/audio cues.
- **Compact Design:** Miniaturize the system for easy integration into office chairs.